2.4.37

Naman Kumar-EE25BTECH11041

6 September,2025

Question

Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $4\hat{\imath} - \hat{\jmath} + 3\hat{k}$

Equation Used

Cross of two vectors=

$$\mathbf{A} \times \mathbf{B} = \begin{pmatrix} |A_{23} & B_{23}| \\ |A_{31} & B_{31}| \\ |A_{12} & B_{12}| \end{pmatrix} \tag{1}$$

And Magnitue or Length of any vecotr is,

$$\|\mathbf{A}\| = \sqrt{(\mathbf{A})^T(\mathbf{A})} \tag{2}$$

Solution

Given,

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} \tag{3}$$

By putting values in the equation

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} = \begin{pmatrix} -1 \times 3 - 2 \times -1 \\ 2 \times 4 - 2 \times 3 \\ 2 \times -1 - 4 \times -1 \end{pmatrix} \tag{4}$$

$$\mathbf{C} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \tag{5}$$

Solution

Unit Vector is

$$\hat{C} = \frac{\mathbf{C}}{\|C\|} \tag{6}$$

$$\hat{C} = \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \tag{7}$$

Vector in the direction of $\pm\mathbf{C}$ with magnitude 6,

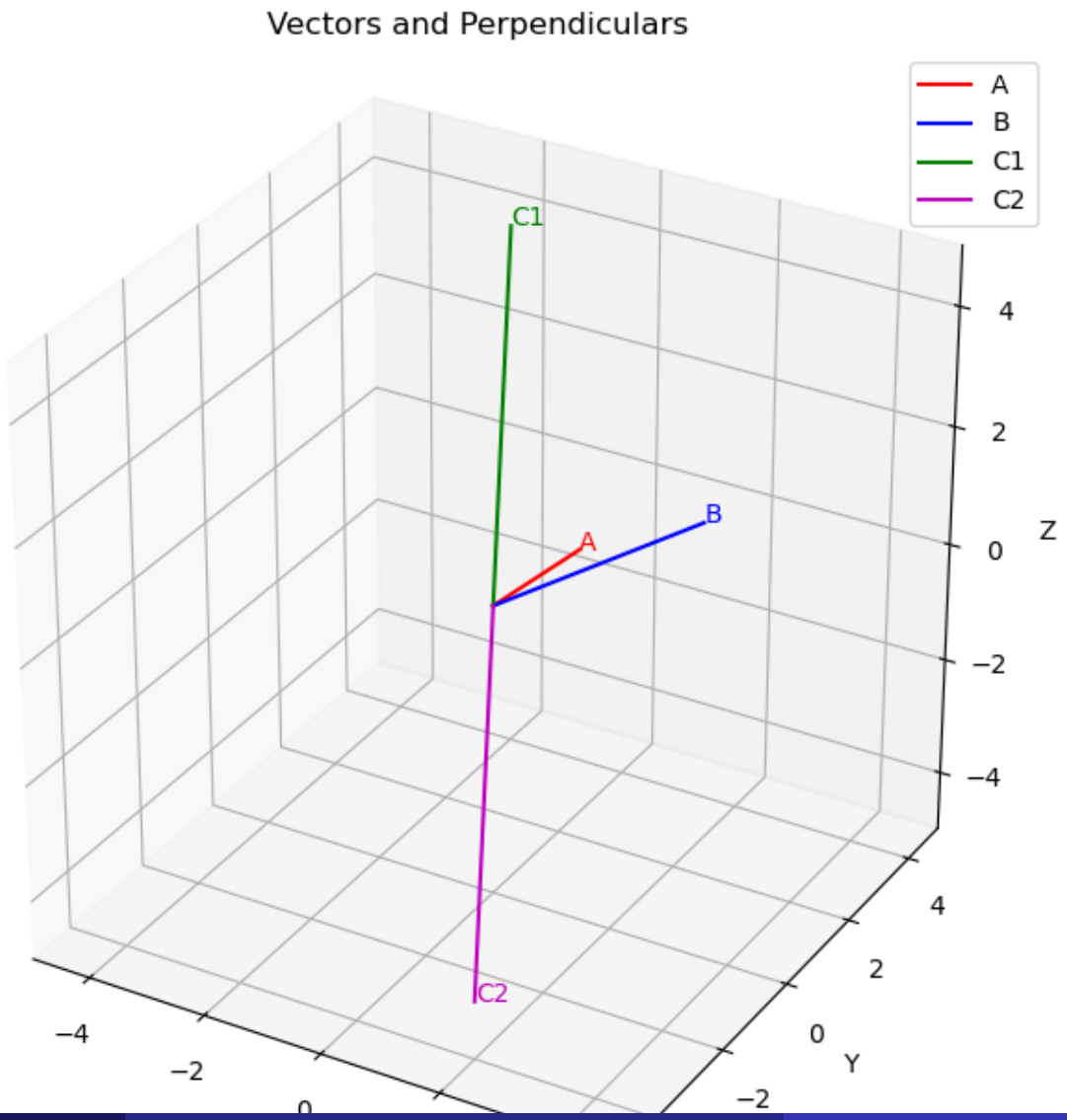
$$\mathbf{D} = \text{Magnitude} \times \text{Unit Vector} \tag{8}$$

$$\mathbf{D} = \pm 6 \times \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \tag{9}$$

Therefore final vectors are

$$C_1 = \begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}, C_2 = \begin{pmatrix} 2 \\ -4 \\ -4 \end{pmatrix} \tag{10}$$

Figure


Vectors and Perpendiculars
A
B
C1
C2
C1
A
B
C2
4
2
0
−2
−4
Z
4
2
0
−2
Y
−4
−2
0

C code

```c
#include <math.h>

// Compute cross product of two 3D vectors
void cross_product(double *a, double *b, double *result) {
    result[0] = a[1]*b[2] - a[2]*b[1];
    result[1] = a[2]*b[0] - a[0]*b[2];
    result[2] = a[0]*b[1] - a[1]*b[0];
}

// Compute magnitude of a 3D vector
double magnitude(double *v) {
    return sqrt(v[0]*v[0] + v[1]*v[1] + v[2]*v[2]);
}
```

C code

```c
// Given a, b, and desired magnitude, compute two perpendicular
    vectors of that magnitude
// Output: v1[3], v2[3]
void perpendicular_vectors(double *a, double *b, double mag,
    double *v1, double *v2) {
    double c[3];
    cross_product(a, b, c);

    double norm_c = magnitude(c);

    if (norm_c == 0.0) {
        // Parallel vectors  cross product is zero
        v1[0] = v1[1] = v1[2] = 0.0;
        v2[0] = v2[1] = v2[2] = 0.0;
        return;
    }
```

C code

```
    // Normalize c
    double c_hat[3];
    c_hat[0] = c[0] / norm_c;
    c_hat[1] = c[1] / norm_c;
    c_hat[2] = c[2] / norm_c;

    // Scale to desired magnitude
    v1[0] = mag * c_hat[0];
    v1[1] = mag * c_hat[1];
    v1[2] = mag * c_hat[2];

    v2[0] = -v1[0];
    v2[1] = -v1[1];
    v2[2] = -v1[2];
}
```

Python Code through SO file

```python
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load C shared object
lib = ctypes.CDLL("./main.so")

# Define argument and return types
lib.perpendicular_vectors.argtypes = [
    ctypes.POINTER(ctypes.c_double), # a
    ctypes.POINTER(ctypes.c_double), # b
    ctypes.c_double, # magnitude
    ctypes.POINTER(ctypes.c_double), # v1
    ctypes.POINTER(ctypes.c_double) # v2
]
lib.perpendicular_vectors.restype = None
```

Python Code through SO file

```
# Given vectors
a = np.array([2.0, -1.0, 2.0], dtype=np.double)
b = np.array([4.0, -1.0, 3.0], dtype=np.double)
magnitude = 6.0

# Prepare outputs
v1 = np.zeros(3, dtype=np.double)
v2 = np.zeros(3, dtype=np.double)

# Call C function
lib.perpendicular_vectors(
    a.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    b.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    magnitude,
    v1.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    v2.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
)
```

Python Code through SO file

```
print(f"Vector a: {a}")
print(f"Vector b: {b}")
print(f"First perpendicular vector v1 (|v1|={np.linalg.norm(v1)
    :.2f}): {v1}")
print(f"Second perpendicular vector v2 (|v2|={np.linalg.norm(v2)
    :.2f}): {v2}")

# --- Plotting ---
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
origin = [0, 0, 0]

ax.plot([0,a[0]], [0,a[1]], [0,a[2]], color='r', label='A')
ax.plot([0,b[0]], [0,b[1]], [0,b[2]], color='b', label='B')
```

Python Code through SO file

```
ax.plot([0,v1[0]], [0,v1[1]], [0,v1[2]], color='g', label='V1')
ax.plot([0,v2[0]], [0,v2[1]], [0,v2[2]], color='m', label='V2')

ax.text(a[0], a[1], a[2], 'A', color='r')
ax.text(b[0], b[1], b[2], 'B', color='b')
ax.text(v1[0], v1[1], v1[2], 'C1', color='g')
ax.text(v2[0], v2[1], v2[2], 'C2', color='m')

max_val = np.max(np.abs(np.vstack((a, b, v1, v2)))) * 1.2
ax.set_xlim([-max_val, max_val])
ax.set_ylim([-max_val, max_val])
ax.set_zlim([-max_val, max_val])
ax.set_xlabel('X')
```

Python Code through SO file

```
ax.set_ylabel('Y')
ax.set_zlabel('Z')
ax.set_title('Vectors and Perpendiculars')
ax.legend()
ax.grid(True)
ax.set_box_aspect([1,1,1])

plt.savefig('vector_plot.png')

plt.show()
```

Direct Python Code

```
# Inspired by code from GVV Sharma
# September 5, 2024
# released under GNU GPL
# Find two vectors of magnitude 6 perpendicular to two given
    vectors and plot them.

import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Given vectors
a = np.array([2, -1, 2])
b = np.array([4, -1, 3])

# To find a vector perpendicular to both a and b, we compute
    their cross product.
c = np.cross(a, b)
```

Direct Python Code

```
print(f"Vector 'a': {a}")
print(f"Vector 'b': {b}")
print(f"Vector perpendicular to a and b (a x b = c): {c}")

# Now, we need a vector of magnitude 6 in the direction of c.
# First, find the unit vector in the direction of c.
# Magnitude of c
norm_c = np.linalg.norm(c)
# Unit vector c_hat
c_hat = c / norm_c

print(f"Magnitude of c: {norm_c:.2f}")
print(f"Unit vector in the direction of c: {c_hat}")
```

Direct Python Code

```python
# Define the desired magnitude
magnitude = 6

# Calculate the final vector v and its opposite
v1 = magnitude * c_hat
v2 = -v1

print(f"The first required vector 'v1' of magnitude {magnitude}
    is: {v1}")
print(f"The second required vector 'v2' of magnitude {magnitude}
    is: {v2}")
print(f"Verification: Magnitude of v1 is {np.linalg.norm(v1):.2f}
    ")
print(f"Verification: Magnitude of v2 is {np.linalg.norm(v2):.2f}
    ")
```

Direct Python Code

```
# --- Plotting the vectors ---
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# Origin point
origin = [0, 0, 0]

# Plotting the vectors as lines from origin
ax.plot([origin[0], a[0]], [origin[1], a[1]], [origin[2], a[2]],
    color='r', label='A = [2, -1, 2]')
ax.plot([origin[0], b[0]], [origin[1], b[1]], [origin[2], b[2]],
    color='b', label='B = [4, -1, 3]')
ax.plot([origin[0], v1[0]], [origin[1], v1[1]], [origin[2], v1
    [2]], color='g', label=f'Result C1 = [{v1[0]:.0f}, {v1[1]:.0f
    }, {v1[2]:.0f}]')
```

Direct Python Code

```
ax.plot([origin[0], v2[0]], [origin[1], v2[1]], [origin[2], v2
    [2]], color='m', label=f'Result C2 = [{v2[0]:.0f}, {v2[1]:.0f
    }, {v2[2]:.0f}]')

# Adding text labels at the end of each vector
ax.text(a[0]*1.1, a[1]*1.1, a[2]*1.1, 'A', color='r', fontsize
    =12)
ax.text(b[0]*1.1, b[1]*1.1, b[2]*1.1, 'B', color='b', fontsize
    =12)
ax.text(v1[0]*1.1, v1[1]*1.1, v1[2]*1.1, 'C1', color='g',
    fontsize=12)
ax.text(v2[0]*1.1, v2[1]*1.1, v2[2]*1.1, 'C2', color='m',
    fontsize=12)
```

Direct Python Code

```
# Setting the plot limits to be symmetric and encompass all
    vectors
max_val = np.max(np.abs(np.vstack((a, b, v1)))) * 1.2
ax.set_xlim([-max_val, max_val])
ax.set_ylim([-max_val, max_val])
ax.set_zlim([-max_val, max_val])

# Adding labels and title
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
ax.set_zlabel('Z axis')
ax.set_title('Perpendicular Vector Visualization')
ax.legend()
ax.grid(True)
```

Direct Python Code

```
# Change the viewing angle (elevation, azimuth)
ax.view_init(elev=25, azim=-45)

# To make the aspect ratio equal
ax.set_box_aspect([1,1,1])

# Save the figure as a PNG file
plt.savefig('vector_plot.png')

# Show the plot
plt.show()
```